FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 2

TUTORIAL MS AZURE

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GROUP 2

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Project Overview

Nowadays, people face challenges in gaining faster insights and generating meaningful reports when customer and sales data reside in different systems. Traditional methods of reporting usually take time, require manual work and do not allow interaction between them. This situation becomes challenging for businesses to understand the customer behaviour and make timely decisions based on their data.

This project aims to resolve these issues through the end-to-end implementation of the data engineering pipeline in Microsoft Azure. The goal is automation for extracting data from on-premises SQL Server, cleaning the data, and delivering it as insights in a Power BI dashboard. This allows users to explore and analyse data more efficiently and accessible.

This project manages the whole data pipeline including the data ingestion, storage, transformation and visualization with basic automation and monitoring to simulate a real-world environment. Nevertheless, the project is confined to structured data and does not require real-time streaming or artificial intelligence features. Overall, it demonstrates how cloud-based tools can be used to create an organized and automated data pipeline for informed decision-making.

Description on the Solution

The solution is designed as an end-to-end data pipeline that transfers data from the source system to a final reporting dashboard through various structured steps. The workflow transforms raw data into meaningful information that is analysis-ready for business reporting.

1. Data Ingestion
   An on-premises SQL server database is used for the extraction of data to the cloud. In this step, source data will centralize and available for further processing in the pipeline. Creating a copy of the data in the cloud for analysis also helps to reduce reliance on the on-prem solution.

2. Data Storage
   The Bronze layer stores data in its raw format and is unfiltered. The first landing zone is where the complete source data is kept as it is. It is crucial to enable traceability and document recovery if any issues occur in later stages.

3. Data Transformation
   Data undergoes processing through numerous stages of transformation. In the Silver layer, data is cleaned by filling missing values, format correction and consistency check. The Gold layer refers to the refined and processed data sets created by applying business logic and aggregations to reporting data. The use of layers guarantees data accuracy and the isolation of processing steps.

4. Data Warehousing

   After processing, the data is arranged into structures views to support querying and analysis. These views enable reporting and thus simplify access to data that is business ready. This layer boosts performance when querying large data analytics.

5. Data Visualization

   The final cleaned dataset is used to build an interactive dashboard where users can interact with the insights using filters and charts. The dashboard enables stakeholder to compare important metrics and trends without having know technical skills. This enhances decision-making and availability of data.

6. Automation and Monitoring

   The pipeline can be set to operate automatically and at regular intervals to keep processing the data. The execution of the pipeline is monitored, the failures are detected or data is loaded successfully can be shown with the help of monitoring tools. This saves manual labour and enhances reliability of the system.

Technology Use

This tutorial uses several Microsoft Azure tools to build a system that moves, cleans, analyses data and at the end ready for visualisation. Each tool has a specific job in the pipeline to make sure the data is reliable and useful.

Tools	Purpose
SQL Server (On-Premises)	The starting point for our project's system. It is a database sitting on a local computer that holds all the original sales information. We use it to represent our main data source that need to move the data into the modern cloud.
Azure Data Factory (ADF)	Acts as the data orchestrator and ingestion engine. By using the Self-hosted Integration Runtime (SHIR), ADF securely bridges the gap between the local SQL Server and the Azure cloud, allowing for automated and scheduled data movement.
Azure Data Lake Storage Gen2	Serves as the central storage repository. It supports the Medallion Architecture, providing separate containers for different data states (Bronze, Silver, and Gold) which ensures better data governance and traceability throughout the process.
Azure Databricks	It is used to perform data transformation. It works perfectly with ADLS Gen2 and can handle massive amounts of Big Data much faster than a regular computer. We write our instructions in PySpark notebooks, which use the Python language.
Azure Synapse Analytics	It is used to create SQL Views, which makes the data organized and easy for tools like Power BI to read without moving the data out from the storage again.
Azure Key Vault	It is used as our secret manager. Instead of writing down passwords or keys in our code, we hide them inside this key vault. This keeps our project secure from hackers.
MS Power BI	The final visualization layer. It connects to the Synapse serving layer to turn raw business data into actionable insights

	through interactive dashboards, allowing stakeholders to make data-driven decisions.

Data Pipeline Architecture

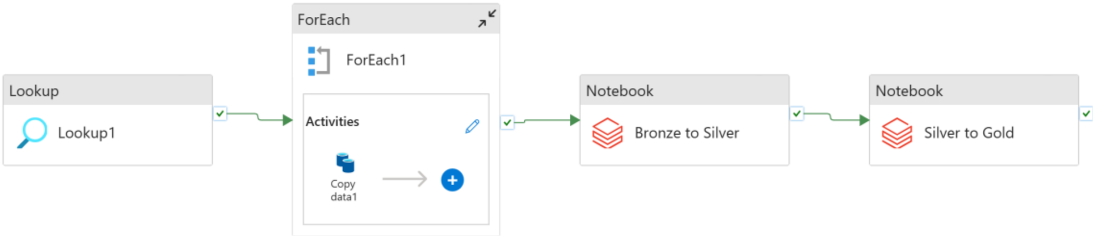

Figure 1: Data Pipeline of Tutorial

Stage 1: Data Ingestion (From Source to Bronze)

The journey starts with the On-Prem SQL Database. Since we have many tables to move, we use Azure Data Factory to automate the work. First, at the Lookup stage, the pipeline gets a list of all the table names from the SQL Server. In ForEach stage, it loops through that list one by one. For every table, it copies and pastes the data into our Bronze Layer which are still the raw storage as Parquet files. This keeps an exact copy of the original data in the cloud.

Stage 2: Data Transformation (From Bronze to Silver)

Once the raw data is in the Bronze layer, we use Azure Databricks to transform and clean it up. We fix things like complex dates, remove empty rows and make sure the columns have right names. The cleaned data is saved into the Silver Layer. This layer is much easier for developers to work with because the "bad" condition data has been removed.

Stage 3: Data Enrichment (From Silver to Gold)

Now that the data is clean, we use Azure Databricks again for the final preparation. Instead of doing it manually, we can write a script to automatically list all folders we have and standardize the column names into a cleaner format. This final step ensures that all the information in the Gold Layer is perfectly formatted and easy for the business team to use or read in their final reports.

Stage 4: Visualisation & Analytics (Gold to Power BI)

The final step is making the data easy to read for the business users. We can use Azure Synapse to create "Views" of our data. This allows us to use SQL to investigate the Gold files without moving the dataset out again back and forth. Finally, Power BI connects to these Synapse views to create the interactive charts and graphs. This is where the raw numbers finally become useful business insights.

MS Power BI dashboard

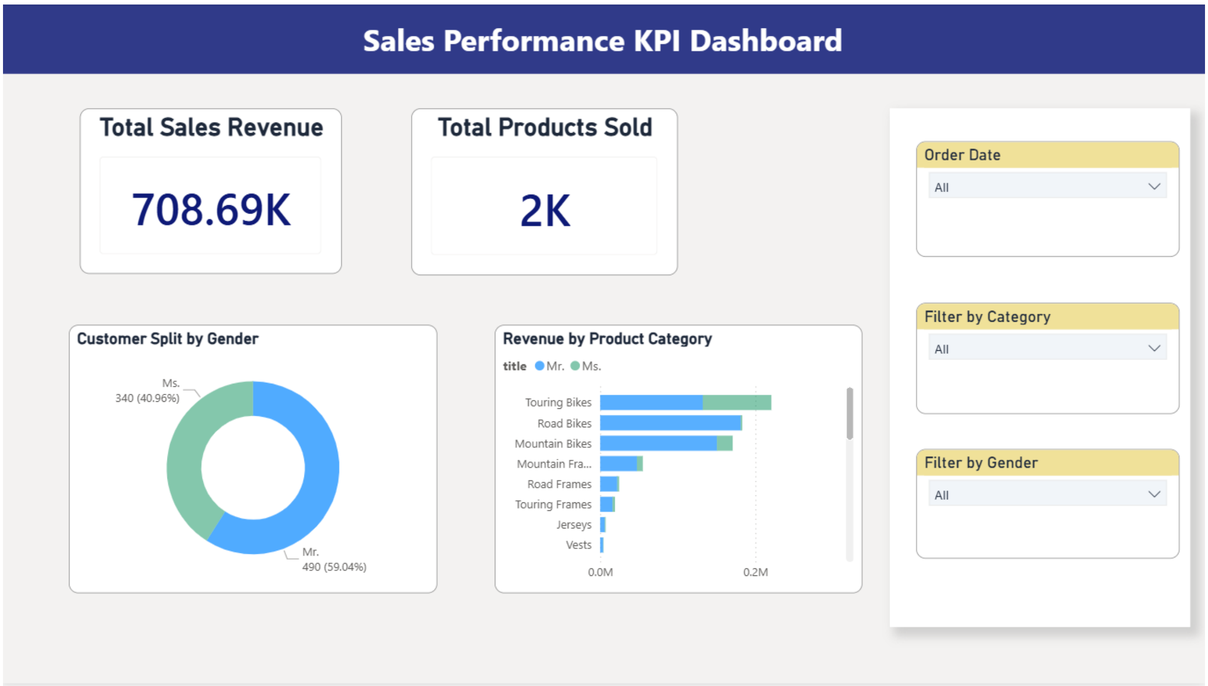

Figure 2: Power BI Dashboard

The final dashboard is designed to be user-friendly and enables managers to easily understand sales performance and customer behaviour. On the right side, a control panel in the form of slicers allows users to filter the data by order date, product category, and gender. This interactive feature enhances data exploration by enabling users to focus on specific segments of interest.

At the top of the dashboard, card visuals are used to present key performance indicators (KPIs). These include the total sales revenue of **$708.69K** and the total number of products sold, which is **2,000 units**. The use of cards ensures that important metrics are clearly visible and quickly accessible.

To provide deeper insights, charts are used to visualise patterns and comparisons in the data. A donut chart illustrates the distribution of customers by gender, showing that 59% are male ("Mr.") and 41% are female ("Ms."). Additionally, a bar chart displays the revenue generated by different product categories, such as Touring, Road, and Mountain Bikes. The charts apply a consistent colour scheme, where blue represents male customers and green represents female customers, making it easier to compare purchasing behaviour. These visualisations help identify that male customers contribute more to bike purchases, which can support more targeted marketing strategies.

Reflection

Chau Ying Jia

The whole process of building this end-to-end data pipeline using Azure was an amazing practical exercise for me since I got to learn how to get the raw data from data source until building the Power BI report for visualisation. Even though the tutorial itself took only two and a half hours, I took about three days to complete this task. There are numerous of unexpected connection errors that I encountered during my attempt to do it. I spent quite some time to solve these problems seeking help from Gemini and exploring Azure services. This experience taught me that the most important part of learning is being independent and having the drive to solve the challenges you face. It has inspired me to keep upskilling, as there is always more to learn in the field of data engineering.

Lau Yee Wen

This project gave me great hands-on experience building a complete Azure data pipeline, from extracting local SQL data to creating a Power BI dashboard. My main challenge was following the tutorial video. The video showed a flawless process, but I ran into several unexpected errors and bugs in my own setup. Since the video didn't explain how to fix these issues, I had to spend a lot of extra time exploring Azure, researching the error messages, and troubleshooting on my own. While frustrating at first, figuring out how to fix these unscripted errors was the most educational part of the project and greatly improved my independent problem-solving skills.

Poh Lok Yee

The project made me realize the functionality of a complete data engineering pipeline in more than theoretical terms. Although the tutorial appeared simple and just a few hours, but I encountered a few problems when implementing it, especially the service configuration and pipeline errors, which did not necessarily work as expected and took me few days to complete it. I wasted precious time in debugging issues like data movement failure and minor errors in the workflow that led to the pipeline breaking. These issues helped me to understand that it takes time and many efforts to create a working system. It was an experience that enhanced my problem-solving abilities and demonstrated that true learning occurs when you are solving an unexpected problem rather than following the instructions.

Clockify Logbook Report

Detailed report

23/03/2026 - 24/04/2026

Total: 53:56:58 Billable: 53:56:58 Amount: 0.00 USD

Date	Description	Duration	User
10/04/2026	watch tutorial and debug error Group 2 Section 02 - Watch Youtube Tutorial	05:49:42 14:23:00 - 20:12:42	Poh Lok Yee 0.00 USD
11/04/2026	watch tutorial Group 2 Section 02 - Watch Youtube Tutorial	02:58:29 14:52:00 - 17:50:29	Poh Lok Yee 0.00 USD
11/04/2026	Follow the a tutorial in YouTube on Azure End-To-End Data Engineering Project for Beginners (FREE Account) | SQL DB Tutorial Group 2 Section 02 - Watch Youtube Tutorial	06:00:00 18:00:00 - 00:00:00 +1	lauyeewen 0.00 USD
12/04/2026	Follow the a tutorial in YouTube on Azure End-To-End Data Engineering Project for Beginners (FREE Account) | SQL DB Tutorial Group 2 Section 02 - Watch Youtube Tutorial	08:00:00 12:00:00 - 20:00:00	lauyeewen 0.00 USD
12/04/2026	watch tutorial and debug error Group 2 Section 02 - Watch Youtube Tutorial	04:34:13 16:18:00 - 20:52:13	Poh Lok Yee 0.00 USD
13/04/2026	watch youtube and flw Group 2 Section 02 - Watch Youtube Tutorial	05:07:46 11:48:03 - 16:55:49	Chau Ying Jia 0.00 USD
13/04/2026	Coordinated with members to assign specific roles and responsibilities Group 2 Section 02 - Distribute Task	00:29:00 20:45:00 - 21:14:00	lauyeewen 0.00 USD
13/04/2026	Discuss and distribute task for all members for the report Group 2 Section 02 - Distribute Task	00:29:36 20:45:00 - 21:14:36	Poh Lok Yee 0.00 USD
13/04/2026	Write report overview Group 2 Section 02 - Write Report: Overview	00:29:33 20:45:00 - 21:14:33	Poh Lok Yee 0.00 USD
14/04/2026	Write report description of the solution Group 2 Section 02 - Write Report: Description on the Solution	01:21:54 13:28:00 - 14:49:54	Poh Lok Yee 0.00 USD
14/04/2026	continue tutorial video Group 2 Section 02 - Watch Youtube Tutorial	03:18:00 14:29:00 - 17:47:00	Chau Ying Jia 0.00 USD
14/04/2026	Write Reflection Group 2 Section 02 - Write Individual Reflection	00:24:28 14:52:00 - 15:16:28	Poh Lok Yee 0.00 USD
14/04/2026	trying to complete the tutorial and learn from the problems and errors i met Group 2 Section 02 - Watch Youtube Tutorial	05:53:00 19:13:00 - 01:06:00 +1	Chau Ying Jia 0.00 USD
15/04/2026	left around 40 minutes trying to complete Group 2 Section 02 - Watch Youtube Tutorial	03:54:00 20:38:00 - 00:32:00 +1	Chau Ying Jia 0.00 USD

16/04/2026	Writing report Group 2 Section 02 - Write Report: Technology used	01:27:00 11:27:00 - 12:54:00	Chau Ying Jia 0.00 USD
16/04/2026	Continue Technology use and proceed to Data Pipeline Architecture Group 2 Section 02 - Write Report: Data Pipeline Architectture	01:46:00 17:36:00 - 19:22:00	Chau Ying Jia 0.00 USD
16/04/2026	Write Reflection Group 2 Section 02 - Write Individual Reflection	00:34:00 19:23:00 - 19:57:00	Chau Ying Jia 0.00 USD
17/04/2026	Enhanced the MS Power BI dashboard to improve user navigation and insights Group 2 Section 02 - Write Report: MSI Power BI Dashboard	00:45:00 19:00:00 - 19:45:00	lauyeewen 0.00 USD
17/04/2026	Write personal contributions, challenges faced during the project, and key learning outcomes for this tutorial Group 2 Section 02 - Write Individual Reflection	00:15:00 20:00:00 - 20:15:00	lauyeewen 0.00 USD
18/04/2026	Check clockify Group 2 Section 02 - Update Clockify logbook	00:05:17 15:16:28 - 15:21:45	Poh Lok Yee 0.00 USD
18/04/2026	Logged all project hours and activities for the week to ensure accurate time tracking and reporting Group 2 Section 02 - Update Clockify logbook	00:15:00 20:00:00 - 20:15:00	lauyeewen 0.00 USD